🧱 Architecture Overview

The platform is organized into modular subsystems that work together to provide reliable solar monitoring.

High-Level Diagram

Collector → FastAPI Backend → SQLite → Dashboard UI → Analytics Engine

Code

Subsystems

Collector

- Polls panel data
- Validates readings
- Handles retries
- Detects offline panels

Backend (FastAPI)

- Data ingestion
- Historical queries
- Daily summaries
- Health checks

Dashboard UI

- Real-time metrics
- Daily charts
- Panel grid
- Alerts

Analytics Engine

- Trends
- Anomaly detection
- Efficiency metrics

Reliability Layer

- Systemd services
- Watchdogs
- Backups
- Log rotation

Data Flow

1. Collector polls panel hardware
2. Sends data to backend ingestion endpoint
3. Backend stores data in SQLite
4. Dashboard queries backend
5. Analytics jobs compute summaries and insights

Configuration System

Centralized YAML/TOML config with:

- Poll intervals
- Panel definitions
- Backup paths
- Alert thresholds